



Wearable Sensor Technology for Detecting and Alerting Dog Owners to Dog Heat Stress



Bhargav Gudapati • Nivedha Natarajan • Kristain Pitshugin • Nikita Neeli • Sophia Rhoads • Manaswi Gorle • Braxton Scarbrough • James Ricci

Introduction and Motivation



We set out to build a custom sensor for a pet dog that records and transmits data metrics needed to detect heat stress incidents.

Design considerations included:

1. Portability – The module should be designed to be as unintrusive as possible to the subject (dog) wearing it to promote seamless adoption of the collar module into the routines of the owner and dog.
2. Data Breadth – The module must support the ability to read and process all the data metrics needed by our heat-stress predictive algorithm.
3. Resilience – Battery life should be sufficient so that charging is infrequent and the device should be able to withstand the daily activity of a domesticated dog.

What this phase delivered:

- Complete sensor-side software.
- Near complete client-side reading and interpretation.
- Sensors initialized and verified for sanity (temp, humidity, IR, accelerometer).
- A compact enclosure concept focused on resilience in outdoor environments.

Background and Improvements

Previously, we have used Kestrel-type meters. These are premade off-the-shelf devices, but they come with some disadvantages:

- Reliable for temperature/humidity/heat-stress, but are meant for handheld use; too large for a dog to wear
- Fixed firmware/hardware with limited access to raw signals or sampling control.

To solve these issues: Build a sensor module from scratch in which we choose the chips and sensors, control sampling and logging, and design a small, low-profile form factor that's practical and unintrusive for dogs.

Approach and Methodology

Platform MCU: XIAO-NRF52840

- Chosen for its integrated Bluetooth LE for power efficient communication
- Included battery input pins
- Capable of ultra low-power and sleep modes

Sensors Used:

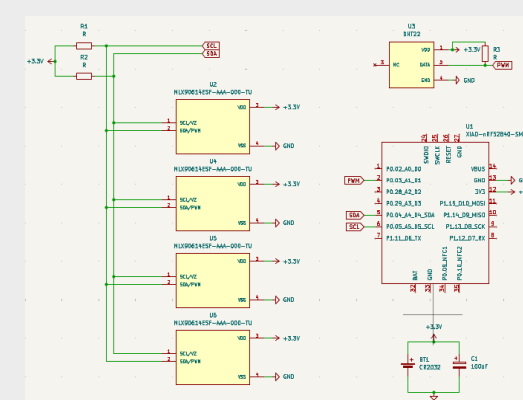
- **IR Thermometer:** MLX90614
 - Integrates Infrared temp readings for accurate skin temp measurement and includes ambient temperature readings.
- **Humidity:** DHT22
 - Low power humidity sensor with accurate readings, and commonly available
- **IMU (accel/gyro):** LSM6DS3TR-C
 - Built into the XIAO MCU.

Power Delivery:

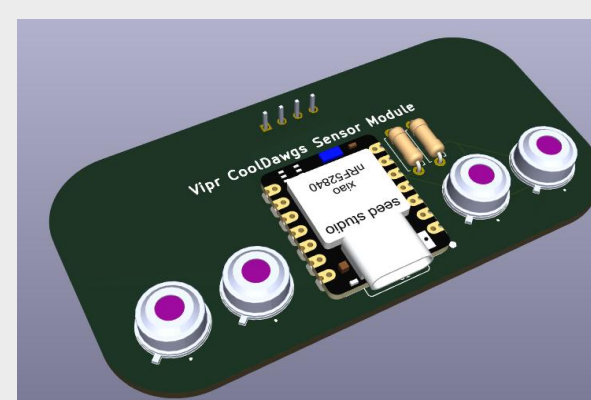
- CR2032 coin-cell selected to eliminate external USB-C port and preserve enclosure waterproof integrity
- Low average current draw allows use of a non-rechargeable primary cell, accelerating prototyping.

Experiments

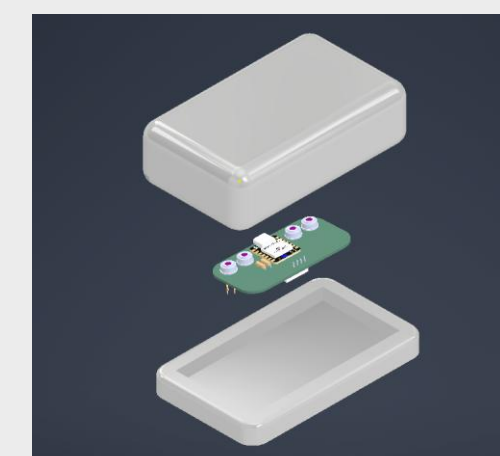
- Dry-runs with console output to confirm sensor readings were sane (no data corruption, malfunction, etc.)
- Test logs with session markers; CSV opens cleanly in a spreadsheet for easy reading.
- Soldered development board using breadboard with all necessary components.
- Quick placement tests (desk/window/outside) to ensure working.
- Battery test: module lasted 4 weeks (28 days) on one charge, polling and transmitting data at 4 times per minute.
- Bluetooth connection – ensured that the Bluetooth connection was possible between our MCU and the companion mobile app that we developed.



(Schematic)



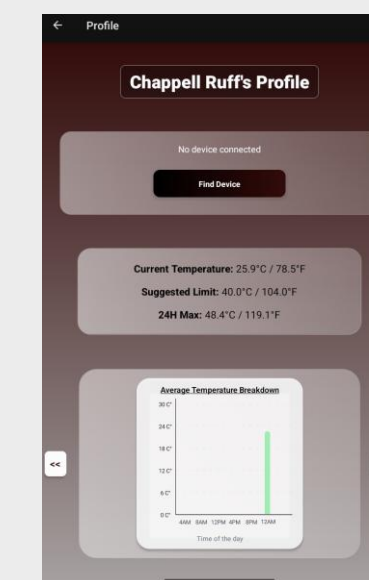
(PCB)



(Enclosure)

Results

- Focus this phase: get software and hardware ready for testing.
- End-to-end read - timestamp - CSV to Flash - export; all core sensors (temperature, IR, humidity, IMU, mag) are integrated and readable in code.
- Soldering (headers + wires on breadboard), reduced wiggle/dropouts; cleaner signals than loose jumpers.
- Mounting/form factor: cuboid shaped enclosure; prototyped with PLA and fit to the dimensions of the sensor.
- Immediate next results to capture (planned): short bench + wear runs, check stability over 30–60 min, start adaptive sampling tests and battery/runtime logging



Summary

- Primary achievement: Got entire code pipeline functioning—sensor reads, time-stamping, and clean CSV logging to Flash, plus export for plotting.
- System status: All core sensors (temperature, IR, humidity, IMU) are integrated and tested in code; mounting concept + soldered connections are in place for stable signals.

Next Steps:

- Add adaptive sampling (low when still, high on motion) and basic battery/runtime tracking.
- Further testing on field with a large variety of dog breeds would prove useful to fine-tuning the sensor's data collection speeds, and sensor positioning on the dog's body. More testing with regards to battery life and ruggedness will allow better enclosure design and fitment.
- The project is code-complete for prototyping and ready to move into data collection and enclosure testing.

References

- Acknowledgment: Guidance and feedback from Dr. Johnsen, Dr. Ramaswamy, and Dr. Barnes
- XIAO documentation - https://wiki.seeedstudio.com/XIAO_BLE/
- Code for testing the link between NRF Connect App and the XIAO nRF52840 - <https://how2electronics.com/send-receive-data-to-mobile-app-with-xiao-ble-nrf52840-sense/>